Hands-On AI for Psychiatry From Prediction to Treatment

A practical, translational workshop connecting psychiatric care, digital phenotyping, multimodal AI, and responsible clinical deployment.

IEEE EMBC 2026
48TH ANNUAL International Conference of the IEEE Engineering in Medicine & Biology Society
26-30 July 2026
Toronto, Canada

EMBC 2026 Workshop | July 26th, 2026 | 8:00 AM - 12:00 PM

Metro Toronto Convention Center

Workshop Website
https://lukazikus.github.io/ai-psychiatry-workshop

WORKSHOP OVERVIEW

This workshop brings together psychiatrists, clinical researchers, and machine learning experts to explore practical and translational applications of artificial intelligence for psychiatric assessment, monitoring, and intervention. The session emphasizes clinically grounded AI approaches that can integrate into real-world psychiatric research and practice, while maintaining strong attention to patient safety, ethics, interpretability, and clinical validation. Through focused presentations, hands-on tutorials, and an interdisciplinary panel discussion, participants will examine how digital phenotyping, multimodal data, and AI can support prediction, treatment planning, and responsible deployment in mental health care.

FORMAT Invited talks, interactive notebook-based activity, and panel discussion

AUDIENCE Clinicians, clinical researchers, biomedical engineers, data scientists, and AI researchers interested in psychiatry and mental health

CORE FOCUS From prediction and monitoring to treatment support, clinical translation, and responsible deployment

INVITED SPEAKERS


Prof. Amir Rahmani
University of California Irvine


Prof. Ervin Sejdic
University of Toronto


Dr. Venkat Bhat
University of Toronto


Dr. Madison Aitken
York University


Matthew Flathers
Harvard University


Christine Shi
Harvard University

WORKSHOP ORGANIZERS


Mai Ali
University of Toronto


Dr. Christopher Lucasius
University of Toronto


Prof. Deepa Kundur
University of Toronto

WORKSHOP SCHEDULE

Session	Topic	Start Time
Opening Remarks	Workshop framing, goals, and clinical motivation **Speaker:** Prof. Deepa Kundur, University of Toronto	8:00 am
Talk 1	From Sensing to Conversation: Personalizing Agentic Chatbots with Wearable and Mobile Data **Speaker:** Prof. Amir Rahmani, University of California Irvine, Centralrive, Livi Agents	8:05 am
Talk 2	Generative AI Applications in Psychiatry **Speaker:** Dr. Venkat Bhat, University of Toronto, CAMH, UHN, UHT	8:40 am
Break	Networking and informal discussion	9:15 am
Talk 3	Matching Methods to Objectives: Building the Right AI Workflow for Psychiatric Research **Speakers:** Matthew Flathers and Christine Shi, Beth Israel Deaconess Medical Center, Harvard University	9:25 am
Talk 4	Translational AI approaches to assess multiple health outcomes **Speaker:** Prof. Ervin Sejdic, University of Toronto, North York General Hospital	10:25 am
Hands-On Activity	Interactive notebook-based AI workflow for psychiatry **Speakers:** Mai Ali, Christopher Lucasius, University of Toronto	11:00 am
Panel Discussion	From Algorithms to Care: Safety, Trust, and Clinical Translation in Psychiatric AI **Panelists:** Prof. Ervin Sejdic, Prof. Amir Rahmani, Dr. Venkat Bhat, Dr. Madison Aitken **Moderators:** Mai Ali, Christopher Lucasius	11:40 am
Closing Remarks	Key takeaways and next steps	12:00 pm

FEATURED THEMES

Digital Phenotyping and Multimodal Sensing
How real-world behavioral and physiological signals from wearables, smartphones, speech, and clinical data can support psychiatric assessment and longitudinal monitoring

AI for Prediction and Treatment Support
How machine learning can help identify risk patterns, forecast symptom changes, and inform personalized approaches to care.

Generative AI and Conversational Agents
How personalized, agentic chatbots may support engagement, psychoeducation, and symptom tracking when paired with appropriate safety guardrails.

Workflow Design for Psychiatric Research
How to match clinical objectives with data sources, preprocessing strategies, model choices, evaluation methods, and interpretation tools.

Responsible Translation
How to address safety, bias, interpretability, privacy, clinical oversight, regulatory considerations, and ethical deployment in mental health AI.

INTENDED TAKEAWAYS

- Clear understanding of practical AI methods for psychiatric research and care.
- Awareness of safeguards needed for responsible use.
- Tools and perspectives to support clinical translation.
- Opportunitieis for interdisciplinary collaboration.

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